

BA0-R systems-modeling prior-art synthesis

Revision 2 - REVIEW-ONLY

Baseline: 89e5486e02d07ff5b97082f73a2e22ac1b4319ae

Status

Includes BA0-R9 OPM alternative-kernel challenge. No external concept is promoted to the DDTA BAE metamodel by this document.

1 Corpus and access boundary

The review combines SRC-0034 and candidate SRC-0039–SRC-0047: NASA modeling guidance, SysML v2, KerML, NASA systems-engineering guidance, ISO/IEC/IEEE 42010, AADL, ArchiMate, Estefan’s MBSE survey, and the reflective Object-Process Methodology (OPM) metamodel.

ISO/IEC/IEEE 42010:2022 and ArchiMate 3.2 remain **ACCESS-LIMITED**. OPM SRC-0047 uses an author-uploaded full-text rendering of the 2005 reflective metamodel; current OPM standard status is corroborated only through official ISO 19450:2024 catalogue metadata.

2 What OPM changes

The first reading ring showed recurring semantic responsibilities. OPM supplies the missing counterexample to the stronger inference that those responsibilities determine one necessary class hierarchy:

```

Element
+-- Entity
|   +-- Thing
|   |   +-- Object
|   |   '-- Process
|   '-- State [owned by Object]
'-- Link
    +-- StructuralLink
    '-- ProceduralLink
  
```

System/environment, physical/informational and scope distinctions can be attributes/classifications. Behavior and flow can be carried by typed procedural links. Multiple diagrams refine or abstract one model. Therefore DDTA must separate **semantic responsibility** from **representation mechanism**.

3 Cross-source semantic-responsibility matrix

Responsibility	NASA/SysML/KerML/AADL convergence	OPM counter-model	DDTA implication
Structure	explicit structural/part/component concepts	Object + structural links	preserve structure; exact root type OPEN
Behavior	function/action/behavior explicit	Process is a first-class Thing	strongly required semantic distinction
Information/resource	item/data/resource-flow concepts	informational essence can classify elements	separate information identity not universally forced
Connection	connector/connection/link concepts	Link is first-class	relation identity strongly evidenced
Interface/endpoint	often explicit	no universal interface root forced	first-class Interface OPEN
Flow/transfer	often explicit with payload	procedural links carry flow	Flow root WEAKENED
State/mode	often explicit	State explicit but owned by Object	State root WEAKENED; semantics remain strong
Boundary/environment	context/boundary/environment explicit	systemic/environmental affiliation on any Element	Boundary root WEAKENED; scope distinction strong
Property/constraint	explicit	attributes, characterization, timing and cardinality	likely property mechanism, not necessarily BAE
View/projection	views/renderings/model kinds	one model, multiple OPDs + textual modality	canonical model != view reinforced
Analysis	AnalysisCase / analysis-oriented models	analysis/simulation historically supported	novelty cannot be generic analysis-on-model

4 Hypothesis status changes

Actor root	REJECTED AS GENERIC ROOT. Actor may survive as a contextual role/specialization.
Asset root	REJECTED AS GENERIC ROOT. Asset is more plausibly a security/relevance interpretation.
Interaction edge	REJECTED AS SUFFICIENT GENERIC RELATION. It collapses connection, interface, flow, action and procedural semantics.
Channel root	FURTHER WEAKENED. Communication path can be connector, bus/component, or physical Link.
Store root	OPEN / WEAKENED. Persistence exists but does not force a universal root.
Contract root	OPEN / WEAKENED. Contract-like semantics may be distributed over interfaces, schemas, constraints and requirements.
Boundary root	OPEN / WEAKENED. The distinction is necessary; a Boundary object is not.
State root	OPEN / WEAKENED. State may be owned by another element.
Flow root	OPEN / WEAKENED. Flow can be typed relation semantics.
InformationObject root	OPEN / WEAKENED AS UNIVERSAL ROOT. Stable payload identity remains a DDTA corpus question.

5 Strong semantic obligations for BA0

A DDTA analyzable representation must be capable of preserving, without semantic collapse:

1. identifiable system referents;
2. behavior/transformation;
3. typed relations/connections;
4. information/resource semantics where relevant;
5. state/mode/condition semantics where relevant;
6. system/environment and responsibility/scope distinctions;
7. properties/constraints relevant to analysis;
8. abstraction/refinement and multiple projections/views from canonical knowledge.

The mechanism remains OPEN: entity, relationship, role, property/classification, owned sub-state, or derived projection.

6 Canonical model versus view

NASA/SysML already supported projection. OPM adds semantically equivalent graphical/textual modalities and refinement/abstraction diagrams from one model. The working direction is therefore:

```
canonical analyzable knowledge
  -> select / abstract / refine / project / render
  -> one or more views
```

A generated diagram or method-specific view is not semantic authority.

7 Boundary result after OPM

system / responsibility scope distinction	STRONGLY EVIDENCED
Boundary as first-class BAE	OPEN / WEAKENED
TrustBoundary	OPEN; likely security overlay

8 Novelty challenge

SysML already has AnalysisCase; AADL/NASA support analysis; OPM supports model analysis/simulation and transformations. DDTA must not claim novelty for having a generic model, analysis envelope, view, or transformation. The candidate contribution remains the governed end-to-end boundary:

```
governed documentation
-> reviewed methodology-neutral analyzable representation
-> replaceable method-owned security-analysis overlay
-> reviewed normalized findings
-> accepted governed-document / requirement change
-> preserved analysis-to-change provenance
-> change-aware re-analysis
```

No reviewed systems-modeling source establishes that entire loop; this is a scoped comparative inference, not proof of novelty.

9 Chapter regression

No Chapter 3 or Chapter 4 reopen trigger is produced by BA0-R9. OPM informs the downstream analysis layer and strengthens the requirement that first-class BAE identity must be justified. ISO 42010 full text, when supplied, should be integrated as a regression source for view/viewpoint/architecture-description claims.

10 BA0 working hypothesis - still OPEN

Working hypothesis

Base Analysis is a canonical, methodology-neutral analyzable representation of governed project/system knowledge, grounded in governed documentation and explicit reviewed analytical additions. It must preserve analysis-relevant semantic distinctions without committing DDTA to one external modeling-language ontology, and support multiple consistent projections/views from one accepted knowledge base.

Non-goals: second narrative hierarchy, SysML subset, OPM subset, DFD, threat-model core, raw LLM/NLP extraction, tool-native schema authority.

11 What BA1 must test

- A. typed-entity-rich kernel
(SysML/NASA-like granularity)
- B. compact entity + typed-relation + properties kernel
(OPM-like pressure case)
- C. DDTA hybrid
first-class identity only where independent lifecycle,
reference, provenance or analysis attachment requires it

The winning representation is the smallest one that preserves required DDTA semantics without ambiguity, semantic loss, generic escape relations, or method-specific contamination.

12 Review gate

BA0-R remains REVIEW-ONLY until user review. ISO 42010 full text may be integrated when available. No BAE metaclass is accepted by this review.